Alejandro Ortiz Perdomo

AI Engineer | Machine Learning Engineer

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PROFESSIONAL SUMMARY

Machine Learning Engineer with end-to-end experience from experimentation to production. Specialized in predictive models (**churn, propensity, NBO**) with quantifiable impact: **Lift 3.0**, **+15%** conversion, **+60%** contactability. Proficient in **Scikit-learn, LightGBM, PyTorch** and **Databricks + MLflow** for MLOps. Extensive **deep learning** experience with Transformers and NLP. Combines strong mathematical foundation (Statistics) with modern software engineering to build reliable and scalable ML pipelines.

WORK EXPERIENCE

AI Engineer - AI Practice

Periferia IT Group
06/2025 - Present

- **AI Agent Platform (SRPA) for the SDLC:** Technical design, conceptualization and documentation of a proprietary *Single-Responsibility Plug-and-play* agent ecosystem that automates the end-to-end development lifecycle (onboarding, user-story generation, IDE assistance, multi-language unit testing with >=**85%** coverage, QA, documentation and technical assistants on LLMs such as **DeepSeek R-1** on **Azure AI Foundry**). Multi-cloud platform (**Azure / AWS / GCP**) used both internally and commercially, delivering **-70%** ramp-up and **60-70%** savings across the development cycle.
- **CV-Job Matching System:** Pipeline with **Azure Document Intelligence** + **AI Search**, deployed as **Dockerized Azure Functions**, processing 500+ CVs/day with **95%** extraction accuracy and vector semantic search reducing screening time by **70%**. Running in production at **10k+ searches/month** with <200ms latency.
- **Pre-sales & Solution Architecture:** Build technical estimates, team sizing and reference architectures for **GenAI, Agentic AI and RAG** solutions for enterprise clients (banking, insurance, retail, energy and telco). Lead technical discovery with stakeholders and position the proprietary agent portfolio in commercial technical proposals.

Freelance AI Engineer - Dental Clinic

Freelance
07/2026 - 08/2026

- **WhatsApp AI Agents:** Designed and built, as sole engineer, two agents with **LangGraph** + **Azure OpenAI**: appointment scheduling synced with **Outlook (Microsoft Graph)**, with LLM intent classification and escalation of complex cases; and clinical-category reminders with *human-in-the-loop* approval.
- **Deterministic guardrails & serverless deployment:** The LLM only interprets; booking is executed by deterministic code with locking to prevent double booking. Deployed on **Azure Functions** (queues with poison-queue handling and deduplication), Table Storage, **Key Vault** with managed identity and Application Insights alerts; **WhatsApp Cloud API** webhook with HMAC verification. Test-driven development with **118 tests**.

Machine Learning Engineer

Global Hitss
09/2023 - 06/2025

- **GenAI & RAG:** Designed and deployed conversational agent with RAG that reduced support tickets by **40%**, processing **10k+ queries/month** with 85% resolution without escalation. Stack: LangChain, Azure OpenAI, FAISS.
- **MLOps on Databricks:** End-to-end data-to-production cycle with **PySpark**, **Delta Lake** and **Unity Catalog**, orchestration with **Workflows**, versioning in the **MLflow Model Registry** and endpoint or batch serving. CI/CD pipeline that cut deployment from 2 weeks to **2 days**, with drift monitoring and automated retraining.
- **Reinforcement Learning:** Productized *Next Best Offer* (NBO) system with A/B testing that increased conversion by **15%** and revenue by **$10k per person**.

- **Churn Model:** Developed predictive model with **Lift of 3.0** and 82% recall in top decile, deployed on Databricks and integrated into the CRM for proactive retention. Impacted **50k+ customers**.

Machine Learning Consultant
Grupo CINTE
08/2023 - 09/2023

- **ML Observability:** Designed comprehensive monitoring methodology for *data/model drift* detection with automated retraining alerts, adopted as the data science team's standard.
- **Optimization:** Audited and optimized LightGBM and Random Forest algorithms, improving computational efficiency and reducing inference times.

Data Scientist
NAE
03/2023 - 07/2023

- **Contactability Model:** Developed **LightGBM** model that increased effective contact rate by **60%** (from 25% to 40%), optimizing time windows and channels. Impacted campaigns with **100k+ contacts/month**.

Data Scientist II
Xolit
05/2022 - 01/2023

- **Payment Propensity Model:** Developed predictive model for payment probability and portfolio agreement, maximizing collection through **automated management priorities** based on propensity scores.
- **Recommendation System:** Implemented optimal contact time-slot recommendation engine per account holder, increasing debt collection management effectiveness.
- **Customer Segmentation:** Designed clustering-based marketing strategy for account holders, enabling personalized communication per segment to optimize collection rates.
- **Production & Monitoring:** Deployed the models as batch jobs orchestrated with **Apache Airflow** and built **Power BI** dashboards tracking collection KPIs and production model performance metrics.

EDUCATION

Master in Software Engineering
Universidad de los Andes
2025 - In Progress

B.S. in Statistics
Universidad del Valle
2011 - 2016

- Data Science Diploma (DS4A) - Correlation One, 2022
- Deep Learning Diploma - Univ. Autónoma Occidente, 2023

CERTIFICATIONS

- AI-102: Microsoft Certified Azure AI Engineer Associate - Microsoft, Feb 2026
- NLP Specialization (RAG/Transformers) - Coursera, Jun 2025
- PyTorch Ultimate - Coursera, Mar 2025
- MLOps and Databricks - Netec, Dec 2024
- AWS Cloud Practitioner - Udemy, Feb 2022

TECHNICAL SKILLS

- **GenAI & LLMs:** LangChain (Agents, RAG), LangGraph, Azure OpenAI (GPT-4, Embeddings), Prompt Engineering, Vector DBs (FAISS, Qdrant, Pinecone), LLM Evaluation (RAGAS)
- **Azure AI & Cloud:** Azure AI Search, Document Intelligence, Azure AI Foundry, Azure Functions, Databricks + MLflow (Workflows, Model Registry), Delta Lake, Unity Catalog
- **Google Cloud Platform (GCP):** Vertex AI (Training, Endpoints, Pipelines), Cloud Run, Pub/Sub, Compute Engine (MIG zonales), Cloud Storage, Cloud SQL

- **Amazon Web Services (AWS):** EKS, Lambda, DynamoDB, SNS, SQS, RDS
- **ML/DL Frameworks:** PyTorch, TensorFlow, Scikit-learn, XGBoost, LightGBM, Hugging Face Transformers
- **MLOps & Engineering:** Python, SQL, PySpark, Docker, CI/CD, Apache Airflow, Kubernetes, Terraform, Go, Angular, Kotlin, FastAPI, Streamlit, Git, GitHub Actions

SOFT SKILLS

Effective Communication, Problem Solving, Analytical Thinking, Team Collaboration, Adaptability, Continuous Learning, Time Management, Technical Leadership

LANGUAGES

- Spanish: Native
- English: Professional Proficiency (B2)

PROJECTS

RAG Assistant with Web Scraping and RAGAS Evaluation

github.com/alejortizp/rag-banco
Conversational assistant over a bank's public website: robots.txt-compliant crawler, **Qdrant** indexing, cross-encoder reranking and per-session memory. Evaluated with **RAGAS** (50 questions, independent LLM judge): faithfulness **0.87**; the reranker improved context utilization by **+8 pts** (A/B).
Technologies: Qdrant, Sentence-Transformers, Cross-Encoder, Groq Llama 3.1, RAGAS, Streamlit, Docker

Email Agent with LangChain

github.com/alejortizp/email-agent
Intelligent agent that queries emails via IMAP, classifies with LLM, and generates contextual automatic responses. Reduces response time by **80%**.
Technologies: LangChain, Azure OpenAI, IMAP, Python, Prompt Engineering

NL-to-SQL/Pandas Agent

github.com/alejortizp/nl-to-code
Agent that translates natural language to SQL queries and Pandas operations. Includes code validation, secure sandbox execution, and automatic results visualization.
Technologies: LangChain, SQL, Pandas, Code Interpreter, ReAct Pattern

ML Pipeline with MLflow: 7-Model Benchmark

github.com/alejortizp/MLOps-Pipeline-
Reproducible multivariate regression pipeline comparing **7 models** (Ridge, ElasticNet, Random Forest, Gradient Boosting, SVR, XGBoost, LightGBM) with hyperparameter search (RandomizedSearchCV, 5-fold CV) and **MLflow** tracking. The best model (Gradient Boosting, R^2 **0.84**) cut MAE by **26%** versus the baseline and is serialized as an inference-ready sklearn pipeline.
Technologies: Scikit-learn, XGBoost, LightGBM, MLflow, Pandas, Joblib

MLOps Text Classification System

github.com/alejortizp/mlops-text-classification
MLOps system classifying customer-support messages (public Twitter dataset): **training**, **batch prediction** and **monitoring** pipelines with accuracy-threshold alerts, plus **champion/challenger** retraining that only promotes a new model if it beats the current one in the **MLflow Model Registry**. TF-IDF + CatBoost model with **0.85** accuracy.
Technologies: MLflow, Model Registry, CatBoost, Scikit-learn, NLTK, TF-IDF, Pandas

ACADEMIC PROJECTS

Teaching-Assistant Platform on GCP (Go + Angular)

MISW4204 Cloud Solutions Development - 2026 - Universidad de los Andes
Top contributor (**60%** of commits, team of 4). Migrated the system from Docker Compose to an **autoscaling MIG + Load Balancer** and then to serverless **Cloud Run + Pub/Sub**, with asynchronous AI-generated PDF reports. JMeter load tests: **158 rps** (p95 292 ms) with **0%** infrastructure errors; **84%** (Go) and **93%** (Angular) coverage; SonarQube A rating.

Technologies: Go (Gin), Angular, PostgreSQL, Cloud Run, Pub/Sub, Cloud SQL, OpenAI, JMeter, SonarQube, GitHub Actions

Event-Driven Delivery Marketplace on AWS

MISW4301 Cloud-Native App Development - 2026 - Universidad de los Andes
Led the AWS architecture and IaC for **11 FastAPI microservices** (hexagonal architecture) on **EKS**, with a serverless extension: **SNS -> SQS** with DLQs, **4 container Lambdas** and DynamoDB, all in **Terraform** deployed via GitHub Actions to 5 environments. Fixed cold starts and IAM permissions, reaching **76/76** E2E assertions with zero failures.
Technologies: AWS (EKS, Lambda, SNS, SQS, DynamoDB, RDS), Terraform, Kubernetes, FastAPI, Docker

Microservices Architecture Experiments

MISW4202 Agile Software Architectures - 2026 - Universidad de los Andes
Hypothesis-driven experiments for a booking system: asynchronous payment-anomaly detection in **538 ms** mean vs. a 2 s target (t-test, $p<0.001$) and **100%** Man-in-the-Middle attack detection with zero false positives (RS256 JWT with revocation, RSA-OAEP, integrity hashing, TLS).
Technologies: Flask, Redis Streams, Celery, Nginx, JWT, RSA, SciPy

QR Alarms: from UX Research to Android + Web App

UX Prototyping - MISO - 2026 - Universidad de los Andes
github.com/a-ortizp/alarmas-qr-app
Full product cycle: user research, **3 usability tests** (20 findings), Figma prototype and Material 3 design system (48 components); then a **22-screen** Android app (ML Kit QR scanning, alarms firing with the app closed) and Angular web app, with signed APK releases via GitHub Actions.
Technologies: Figma, Kotlin, Jetpack Compose, CameraX, ML Kit, Angular, GitHub Actions